

Sensor-driven Chatbot for Senior Companionship in US and Japan

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Overview

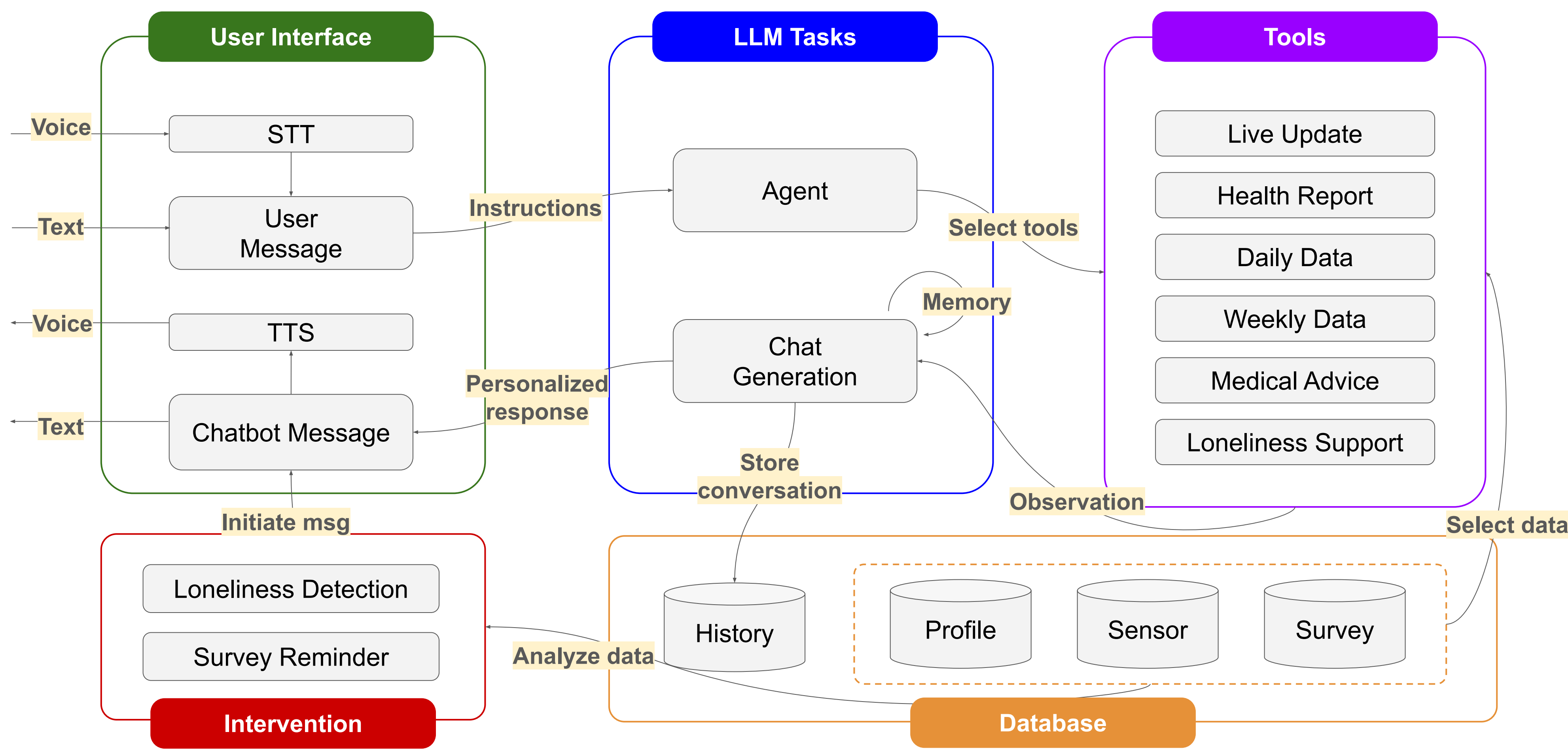
- **Problem Statement**
 - 24/7 personalized support for seniors loneliness using generative chatbot based on passive sensors installed in home
- **Challenges**
 - How to increase the **accessibility** of chatbot to seniors?
 - Surveys
 - How to **design** intuitive UI?
 - Prototypes
 - Speech-to-text/text-to-speech
 - How to address **hallucination** of chatbot?
 - Prompt engineering
 - LLM agent to answer complex questions by activating functions and tools
 - How to address **privacy** concerns?
 - On/off data sharing button
 - Running chatbot in local server
 - Is there a **cultural difference** for chatbot acceptance?
 - Experiments in US and Japan

Chatbot Surveys

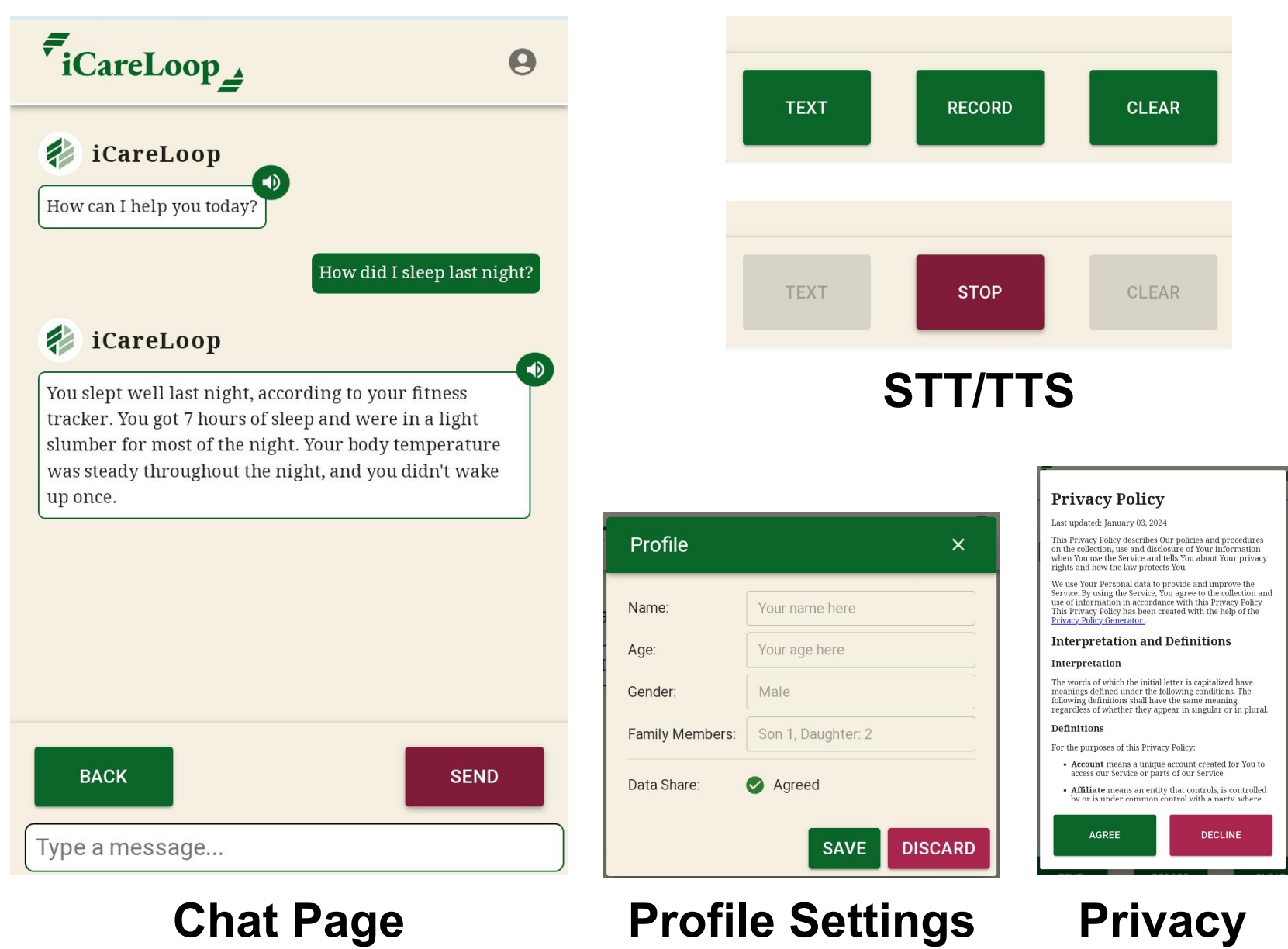


- **Location**
 - 4 senior communities in Philadelphia
- **Collected Surveys**
 - 49 surveys
- **Age Group**
 - 60, 70, 80s
- **Gender**
 - Female: 39
 - Male: 10
- **Ethnicity**
 - Caucasian: 28
 - African American: 19
 - Hispanic: 2
- **Living Arrangements**
 - Living alone: 44
 - Others: 5

Dataflow



Interface Design



Future Work

- **Chatbot Improvements**
 - Develop loneliness detection model
 - Integrate tools and functions
 - Improve agent performance by adding memory and planning
- **Chatbot Experiment**
 - 3 month experiment with passive sensors and chatbot as an intervention targeting seniors over 65
- **Collaboration with Japan**
 - Chatbot surveys
 - Same chatbot experiment
- **Evaluation**
 - Data analysis
 - Post experiment surveys
 - Exit interviews
 - Cross cultural comparison

